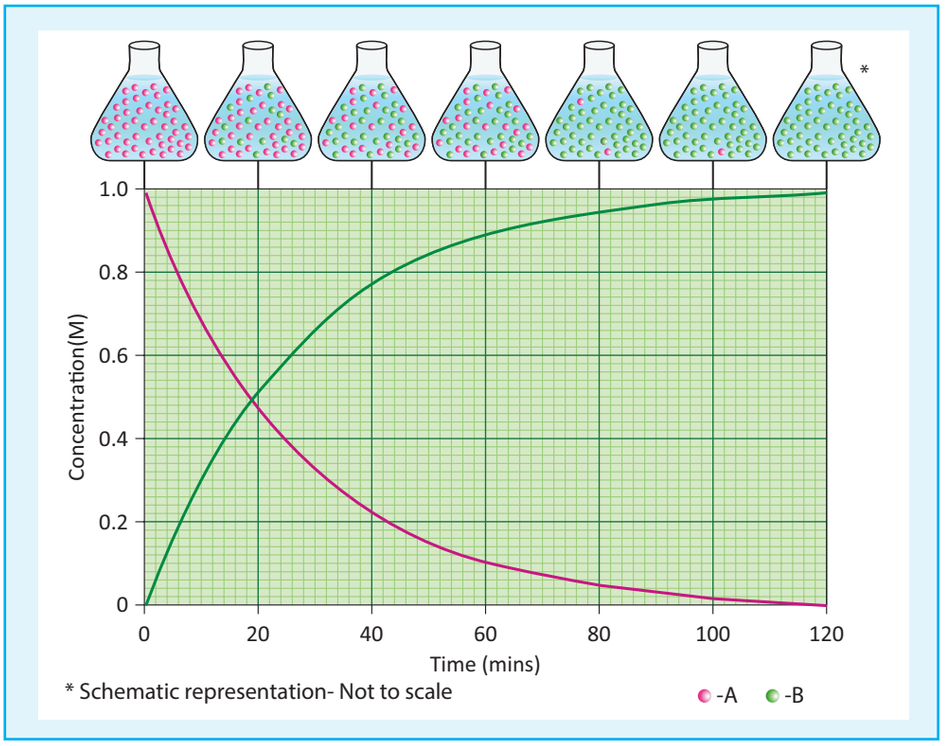


Fig 7.1 change in concentration of A and B for the reaction $A \longrightarrow B$

Unit of rate of a reaction:

$$\text{unit of rate} = \frac{\text{unit of concentration}}{\text{unit of time}}$$

Usually, concentration is expressed in number of moles per litre and time is expressed in seconds and therefore the unit of the rate of a reaction is $mol\ L^{-1}s^{-1}$. Depending upon the nature of the reaction, minute, hour, year etc can also be used.

For a gas phase reaction, the concentration of the gaseous species is usually expressed in terms of their partial pressures and in such cases the unit of reaction rate is $atm\ s^{-1}$.

7.1.1 Stoichiometry and rate of a reaction:

In a reaction $A \longrightarrow B$, the stoichiometry of both reactant and product are same, and hence the rate of disappearance of reactant (A) and the rate of appearance of product (B) are same.

Now, let us consider a different reaction

$$A \longrightarrow 2B$$

In this case, for every mole of A, that disappears two moles of B appear, i.e., the rate of formation of B is twice as fast as the rate of disappearance of A. therefore, the rate of the reaction can be expressed as below

$$\text{Rate} = \frac{+\,d[B]}{dt} = 2\left(\frac{-d[A]}{dt}\right)$$

In other words,

$$\text{Rate} = \frac{-d[A]}{dt} = \frac{1}{2}\frac{d[B]}{dt}$$

For a general reaction, the rate of the reaction is equal to the rate of consumption of a reactant (or formation of a product) divided by its coefficient in the balanced equation

$$xA + yB \longrightarrow lC + mD$$

$$\text{Rate} = \frac{-1}{x}\frac{d[A]}{dt} = \frac{-1}{y}\frac{d[B]}{dt} = \frac{1}{l}\frac{d[C]}{dt} = \frac{1}{m}\frac{d[D]}{dt}$$

XII U7 kinetics - Jerald Folder.indd 206 3/29/2019 6:32:39 PM